

Because electromagnetic waves have mass density equal to their energy density [Section 6.1], they carry physical momentum and move inertially along with the transmitting source. Consequently, rather than expanding as concentric spheres relative to a stationary medium, the points of encounter with successive wavefronts of a moving transmitter form family of nested ellipses.

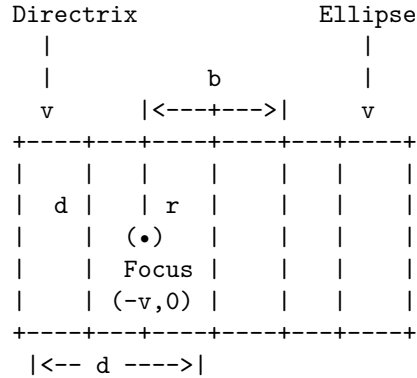
Below is the derivation of the general Doppler shift using the geometry of an ellipse and its directrix, yielding the same expressions as Special Relativity without employing time dilation or length contraction.

Section 5.8: Elliptical Geometry of Wavefront Encounters

Consider a transmitter moving uniformly along the x -axis with constant speed v (in units where $c = 1$, so the eccentricity $e = v$). Due to the inertial motion of the emitted electromagnetic substance, the wavefronts move along with the transmitter.

The resulting points of encounter with these wavefronts form an ellipse: * **Semi-major axis (a)**: $a = 1$ * **Semi-minor axis (b)**: $b = \sqrt{1 - v^2}$ * **Eccentricity (e)**: $e = v$ * **Foci**: Positioned at $\pm ea = \pm v$.

The moving transmitter is situated at the left focus of the ellipse, which serves as the origin of the physical emission.



5.8.1 The Directrix-Focus Relation By the geometric definition of an ellipse, for any point on the curve, the ratio of its distance to a focus (r) to its perpendicular distance to the corresponding directrix (d) is constant and equal to the eccentricity ($e = v$):

$$\frac{r}{d} = e = v \implies r = vd$$

For a stationary transmitter ($v = 0$), the eccentricity is zero, the foci merge at the center, the directrices recede to infinity, and the wavefronts remain circular.

5.8.2 Polar Representation of the Wavefront Using the left focus as the pole for polar coordinates (r, ϑ) , where r is the distance from the transmitter to a point on the ellipse, and ϑ is the angle of the point rotating counterclockwise from the direction of motion (rightwards-pointing), the polar equation of the ellipse is:

$$r(\vartheta) = \frac{p}{1 - e \cos \vartheta}$$

where p is the semi-latus rectum:

$$p = a(1 - e^2) = 1 - v^2$$

Substituting p and $e = v$ into the polar equation:

$$r(\vartheta) = \frac{1 - v^2}{1 - v \cos \vartheta}$$

5.8.3 Wavelength Doppler Shift The wavelength Doppler shift (λ/λ_0) is geometrically determined by comparing the distance from the focus to a point on the ellipse (r) to the length of the semi-minor axis ($b = \sqrt{1 - v^2}$):

$$\frac{\lambda}{\lambda_0} = \frac{r(\vartheta)}{b}$$

Substituting the expressions for $r(\vartheta)$ and b :

$$\frac{\lambda}{\lambda_0} = \frac{1 - v^2}{\sqrt{1 - v^2}(1 - v \cos \vartheta)} = \frac{\sqrt{1 - v^2}}{1 - v \cos \vartheta}$$

Expressing this ratio in terms of the distance d from the point on the ellipse to the directrix (using $r = vd$):

$$\frac{\lambda}{\lambda_0} = \frac{vd}{\sqrt{1 - v^2}}$$

This shows how the directrix geometrically scales the wavelength shift as a function of the angle of encounter.

5.8.4 Frequency (Encounter Rate) Shift Because the frequency ν is determined entirely by the rate of successive wavefront encounters, it is inversely proportional to the wavelength ($\nu \propto 1/\lambda$). Inverting the wavelength ratio yields the general registered frequency:

$$\nu = \nu_0 \left(\frac{1 - v \cos \vartheta}{\sqrt{1 - v^2}} \right)$$

Restoring the explicit speed of light constant c (where $v \rightarrow v/c$):

$$\nu = \nu_0 \left(\frac{1 - \frac{v}{c} \cos \vartheta}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \right)$$

Physical Discussion

- **Longitudinal Recession** ($\vartheta = 0$): The wavefront spacing increases, yielding:

$$\nu = \nu_0 \sqrt{\frac{1 - v/c}{1 + v/c}}$$

- **Longitudinal Approach** ($\vartheta = \pi$): The wavefront spacing decreases, yielding:

$$\nu = \nu_0 \sqrt{\frac{1 + v/c}{1 - v/c}}$$

- **Pure Transverse Shift** ($\vartheta = \pi/2$): The lateral expansion of the ellipse relative to its semi-minor axis yields:

$$\nu = \nu_0 \sqrt{1 - (v/c)^2}$$

This elliptic model recovers the same numerical Doppler shift expressions as Special Relativity. The shift is derived here strictly through classical kinematics, representing the physical momentum of electromagnetic waves moving inertially alongside the transmitter, without requiring the postulation of length contraction or time dilation.